

```
setwd("C:/Users/sadaf/Documents/PPR3")
sf::sf_use_s2(FALSE)
library(sf)
library(dplyr)

# Load
pour_pts <- sf::st_read("data/processed/lake_pour_points.gpkg", quiet = TRUE)
rivers <- sf::st_read("data/processed/HydroRIVERS_main_GB_Chitral.gpkg", quiet
rivers <- sf::st_transform(rivers, 32642)

cat("Pour points:", nrow(pour_pts), "\n")
cat("River segments:", nrow(rivers), "\n\n")

# Function: trace downstream until 50 km cumulative distance
trace_downstream <- function(start_hyriv_id, rivers_sf, max_distance_km = 50)
  current_id <- start_hyriv_id
  visited <- c()
  cumulative_distance <- 0

  while (!is.na(current_id) && current_id != 0 &&
    cumulative_distance < max_distance_km * 1000) {
    if (current_id %in% visited) break
    visited <- c(visited, current_id)

    seg <- rivers_sf |> dplyr::filter(HYRIV_ID == current_id)
    if (nrow(seg) == 0) break

    seg_length_m <- as.numeric(sf::st_length(seg))
    cumulative_distance <- cumulative_distance + seg_length_m

    next_id <- seg$NEXT_DOWN[1]
    current_id <- next_id
  }

  return(list(
    hyriv_ids = visited,
    total_distance_m = cumulative_distance
  ))
}

# Trace for all lakes
cat("Tracing downstream f Start your own conversation ..\n\n")
```